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Registration Number : 23BDS1026

Slot : L23+L24

Course : Exploratory Data Analysis Lab

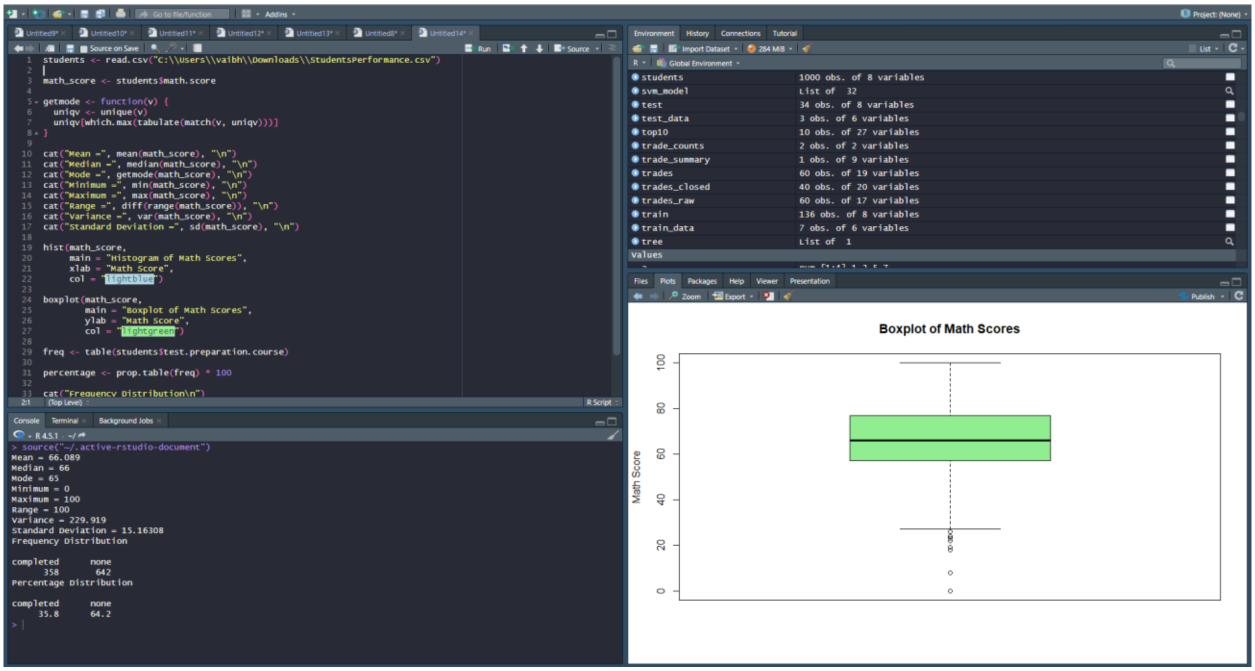
Experiment 6

Code

```
students <- read.csv("C:\\Users\\vaibh\\Downloads\\StudentsPerformance.csv")

math_score <- students$math.score

getmode <- function(v) {
  uniqv <- unique(v)
  uniqv[which.max(tabulate(match(v, uniqv)))]
}

cat("Mean =", mean(math_score), "\n")
cat("Median =", median(math_score), "\n")
cat("Mode =", getmode(math_score), "\n")
cat("Minimum =", min(math_score), "\n")
cat("Maximum =", max(math_score), "\n")
cat("Range =", diff(range(math_score)), "\n")
```

```r
cat("Variance =", var(math_score), "\n")
cat("Standard Deviation =", sd(math_score), "\n")

hist(math_score,
   main = "Histogram of Math Scores",
   xlab = "Math Score",
   col = "lightblue")

boxplot(math_score,
     main = "Boxplot of Math Scores",
     ylab = "Math Score",
     col = "lightgreen")

freq <- table(students$test.preparation.course)

percentage <- prop.table(freq) * 100

cat("Frequency Distribution\n")
print(freq)

cat("Percentage Distribution\n")
print(round(percentage, 2))
```

Output




Code

```
house <- read.csv("C:\\Users\\vaibh\\Downloads\\train.csv")


correlation <- cor(house$GrLivArea,
           house$SalePrice,
           method = "pearson")


cat("Pearson Correlation =", correlation, "\n")


plot(house$GrLivArea,
  house$SalePrice,
  main = "GrLivArea vs SalePrice",
  xlab = "GrLivArea",
  ylab = "SalePrice",
```

```
    pch = 19,
    col = "blue")

abline(lm(SalePrice ~ GrLivArea, data = house),
      col = "red",
      lwd = 2)

boxplot(SalePrice ~ OverallQual,
       data = house,
       main = "OverallQual vs SalePrice",
       xlab = "Overall Quality",
       ylab = "SalePrice",
       col = "lightgreen")

quality_corr <- cor(house$OverallQual,
                  house$SalePrice,
                  method = "pearson")

cat("Correlation between OverallQual and SalePrice =",
   quality_corr, "\n")

if(abs(quality_corr) > abs(correlation))
{
  cat("OverallQual has a stronger relationship with SalePrice.\n")
}else
{
```

```
  cat("GrLivArea has a stronger relationship with SalePrice.\n")
}
```

Output